

MATH 4360 HOMEWORK ASSIGNMENT 1

DUE ON THURSDAY 31 AUGUST 2023

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Consider the polynomials

$$f(x) = 2x^4 + x^3 + 6x^2 + 1 \quad \text{and} \quad g(x) = x^2 + 3x - 1.$$

Find polynomials $q(x)$ and $r(x)$ with $r(x)$ of degree less than 2 such that

$$f(x) = q(x)g(x) + r(x)$$

holds.

- (2) Find polynomials $f(x)$ and $g(x)$ in $\mathbb{Z}_{12}[x]$ of degree 2 such that:

- (a) $f(x)g(x)$ has degree 4.
- (b) $f(x)g(x)$ has degree 3.
- (c) $f(x)g(x)$ has degree 2.
- (d) $f(x)g(x)$ has degree 1.
- (e) $f(x)g(x)$ has degree 0.

- (3) Use the Euclidian algorithm to find the greatest common divisor of

$$f(x) = x^3 + 3x + 1 \quad \text{and} \quad g(x) = x^2 + 4x + 3$$

in $\mathbb{Z}_5[x]$.

- (4) Let $f(x) = x^4 + 5x^2 + 3$.

- (a) Determine the least prime p such that $f(2) = 0$ holds in the field $\mathbb{Z}_p$.
- (b) Factor $f(x)$ as $(x - 2)q(x)$ in $\mathbb{Z}_p[x]$.